

Weekly Update

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1 Main Findings

1.1 DB-Sorter

quick background. DB-sorter uses the sorting characteristic of DB flow to find ground state or excited states of a hamiltonian H using a diagonal matrix D that has an engineered order in its diagonal elements (usually increasing).

Findings:

- This week was mainly spent debugging the warm starts and choosing a better warm start than simple VQE. Finally [VarQITE was implemented](#) and [consistently used](#) for better results. The result was only a sharper drop at the first step while the rest of the steps wasn't any better.
- With some trials and errors, I can see that the base in which D is diagonal has the boldest effect on the energy drop and convergence speed. This introduces a better place for "warm start"s. We can use a variational method as a heuristic to design a proper base for D . See [3.1.1](#)

2 Plan For Next Week(s)

2.1 DB-sorter

- Finding fit basis for D .
- Wrapping up DB-sorter.

3 Details

3.1 DB-sorter

3.1.1 D basis as warm start

Previously, we saw that the energy dip can rely on the basis of D . Looking at the initial equation $\frac{dH}{dt} = [[D, H], H]$ one can assume deciding a basis that maximizes the double bracket would be the way to go. For easier implementation, we try to only maximize the norm of $[D, H]$ on $|k\rangle$ for

k 'th excited state. Which is

$$\begin{aligned}
[D, H] |v_k\rangle &= (DH - HD) |v_k\rangle \\
&= (UMU^*H - HUMU^*)U |k\rangle \\
&= UMU^*HU |k\rangle - HUM |k\rangle \\
&= UMU^*HU |k\rangle - m_0HU |k\rangle \\
&= \sum a_i UMU^*P_iU |k\rangle - a_i m_0 P_iU |k\rangle
\end{aligned}$$

We should be able to heuristicly do sth with it. While I am not sure how to proceed from here, my first idea is to run l parallel variational circuits, one for each Pauli term, with shared parameters defining U .